EE5502
MOS Devices

Long Channel MOSFETs
- Drain Current Models

Zhu Chunxiang
Department of Electrical and Computer Engineering
National University of Singapore
Email: elezhucx@nus.edu.sg
Phone: 6516 8930
Office: E1-08-08

Outline

- Basic MOSFET Operation
- **Drain Current Models**
- Channel Length Modulation
- Threshold Voltage
- Sub-threshold Characteristics
- Body (Substrate Bias) Effect
- MOSFET Channel Mobility

Drain Current Calculation

General Approach:

- Static and dynamic characteristics of a device under the influence of external fields in general can be described by the following three sets of coupled differential equations:

1. Poisson equation: $\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0 \varepsilon_{si}}$

2. Current equation: $J_n = q\mu_n nE + qD_n \nabla n \qquad J_p = q\mu_p pE - qD_p \nabla p$

   Under non-equilibrium, $J_n = -qn\mu_n \nabla \varphi_n \qquad J_p = -qp\mu_p \nabla \varphi_p$

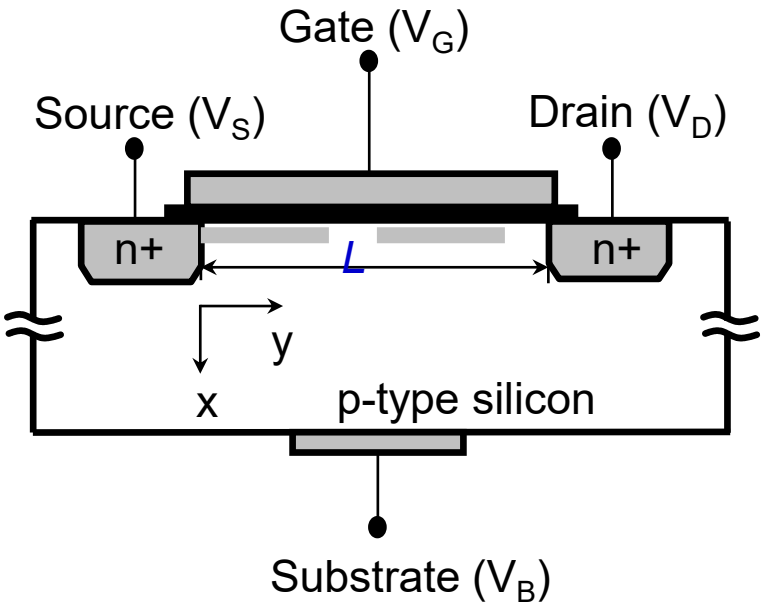
   Where φ_n and φ_p are the electron and hole quasi-Fermi potentials

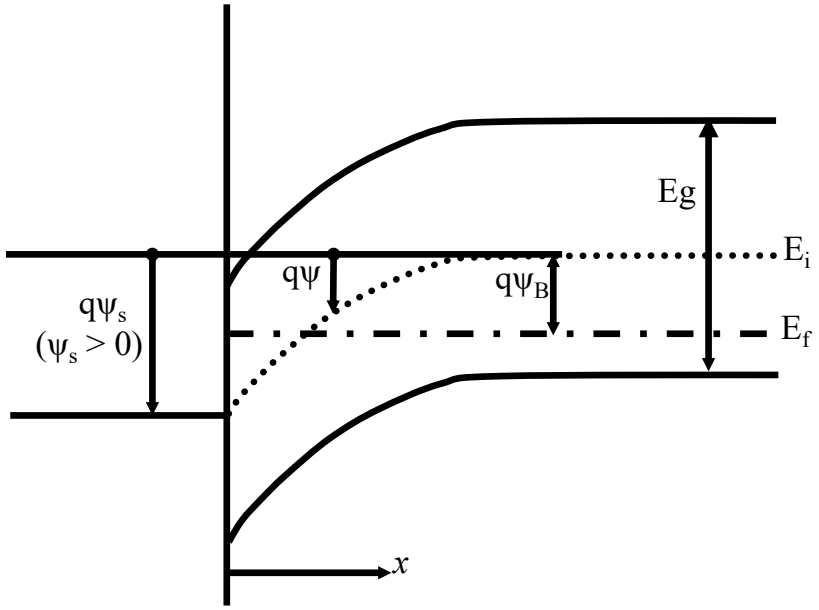
3. Continuity equation:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \bullet J_n - R_n + G_n \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \bullet J_p - R_p + G_p$$

- In general, 3-D problem; mostly, 2-D problems.
- Numerical techniques, like Sentaurus, MEDICI, PISCES, MINIMOS, …

Two Potentials $V(y)$ and $\psi(x)$





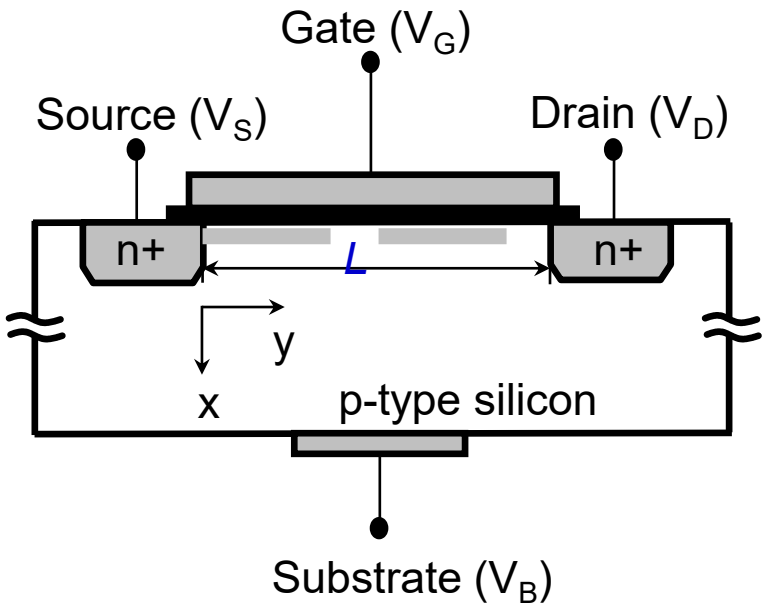
V(y) : The electron quasi-Fermi level potential at a point y along the channel with respect to the Fermi potential of the n+ source.

$\psi(x)$ is defined as

$$-(E_i(x) - E_i(x = \infty))/q$$

It reflects the band bending.

Notes on V(y)



- Since there is no bias between the source and substrate, the Fermi potential of the source is the same as that of the bulk substrate.
- So, in terms of the channel-to-substrate diode, V(y) plays the same role as the reverse bias V_R on MOS gated diode under non-equilibrium.
- At the drain end of the channel, $V(y=L)=V_{DS}$; while at source side, $V(y=0)=0$.

Inversion Charge Density

- The electron concentration at any point (x,y) is given by

$$n(x,y) = \frac{n_i^2}{N_A} e^{q(\psi - V)/kT}$$

- Using the similar approach as MOS-C, the electric field can be obtained as

$$E^2(x,y) = \left(\frac{d\psi}{dx}\right)^2 = \frac{2kTN_A}{\varepsilon_{Si}}\left[\left(e^{-q\psi/kT} + \frac{q\psi}{kT} - 1\right) + \frac{n_i^2}{N_A^2}\left(e^{-qV/kT}\left(e^{q\psi/kT} - 1\right) - \frac{q\psi}{kT}\right)\right]$$

- The condition for surface inversion becomes $\psi(0,y) = V(y) + 2\psi_B$
- The maximum depletion layer width is

$$W_{dm}(y) = \sqrt{\frac{2\varepsilon_{Si}[V(y) + 2\psi_B]}{qN_A}}$$

Recall from MOS-C

$$n(x) = n_i e^{[E_f - E_i(x)]/kT} = n_i e^{q[\psi_i(x) - \psi_f]/kT}$$
$$= n_i e^{q[\psi(x) - \psi_B]/kT} = \frac{n_i^2}{p_0} e^{q\psi(x)/kT}$$

$$\left(\frac{d\psi}{dx}\right)^2 = \xi^2(x) = \frac{2kTN_a}{\varepsilon_{si}}\left[\left(e^{-\frac{q\psi}{kT}} + \frac{q\psi}{kT} - 1\right) + \frac{n_i^2}{N_a^2}\left(e^{\frac{q\psi}{kT}} - \frac{q\psi}{kT} - 1\right)\right]$$

Drain Current Calculation

Assumptions Made

1. Gradual Channel Approximation
2. Quasi-Fermi level in bulk is constant from surface into bulk
   - Current flow is only between source and drain terminals
3. Electrons are the only charge carrier (if considering NMOS)
   - If considering PMOS, holes are the only charge carrier
4. Recombination-Generation is negligible
5. Surface mobility is constant

Assumption #1: Gradual Channel Approximation

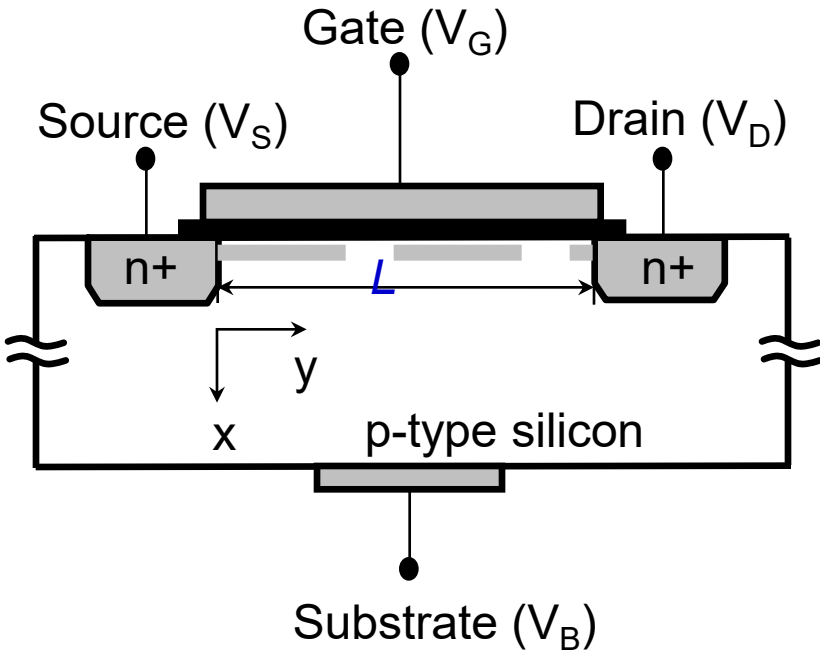
- GCA: the variation of the electric field in the y-direction (along the channel) is much less than the corresponding variation of the field in the x-direction

$$\left|\frac{dE_y}{dy}\right| << \left|\frac{dE_x}{dx}\right|$$

- Importance: reduce 2-D problem into 1-D problem
- Poisson equation is now reduced to:

$$\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_0 \varepsilon_{Si}}$$

- Restrictions
  (1) The GCA is valid for most of channel region
  (2) Not valid beyond the pinch-off point



Assumption 2: Hole current can be neglected

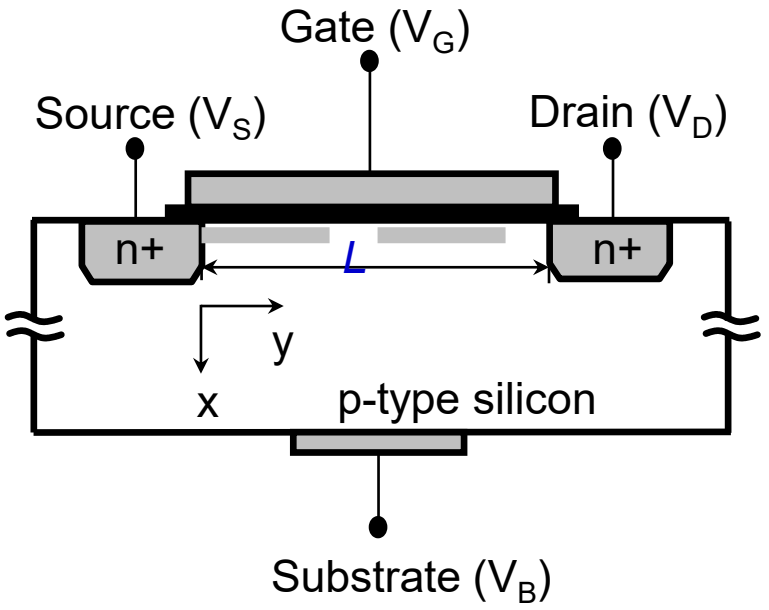
- Importance: only need to consider current density J_n
- Restrictions
  - (1) Valid for normal operation
  - (2) Not valid when device enters breakdown region (will be discussed later)

Assumption 3: Neglect recombination/generation

- Importance: $R_n=G_n=0$
- If only consider static characteristics of the device, the continuity equation reduces to

$$\nabla \cdot J_n = 0$$

- Physical meaning: Total drain current I_{DS} is constant at any point along the channel.



Assumption 4:
Current flows in the y-direction only

Note:
V(y) is the electron quasi-Fermi potential at location y w.r.t. Fermi potential of n+ source

- Importance: $dE_{Fn}/dx=0$, thus the current density J_n reduces to:

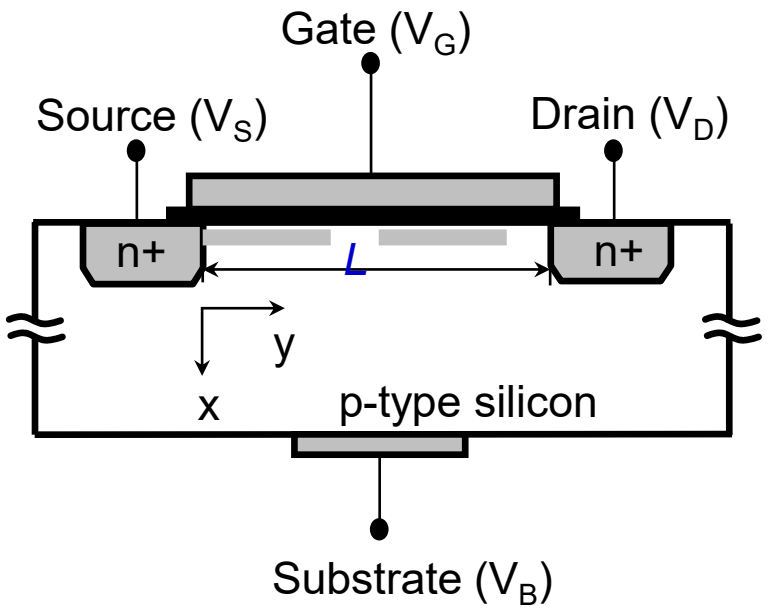
$$J_n(x,y) = -q\mu_n n(x,y)\frac{dV(y)}{dy}$$

- Multiplying above equation with the channel width W and integrating over the depth of the inversion layer. The integration is carried out from x=0 (surface) to x=x_i (bottom of inversion layer where ψ=ψ_B). So, the drain current

$$I_{DS}(y) = qW\int_0^{x_i}\mu_n n(x,y)\frac{dV(y)}{dy}dx$$

Since dV(y)/dy only depends on y, it can be moved outside of integration term.

- Assumption 4 is examined in 2-D simulator with excellent accuracy.



Note: there is a sign change, since we define I_{DS}>0 to be the drain-to-source current in the – y direction.

Assumption 5:
Surface mobility μ_n is constant

$$I_{DS}(y) = qW\int_0^{x_i} \mu_n n(x,y)\frac{dV(y)}{dy}dx$$

- Importance: μ_n can be taken out of integral, using μ_{eff} replacing μ_n
- Introducing the inversion charge per unit gate area, $Q_i(y)$

$$Q_i(y) = -q\int_0^{x_i} n(x,y)dx$$

Then, the drain current can be calculated by:

$$I_{DS}(y) = -\mu_{eff} W \frac{dV}{dy} Q_i(y)$$

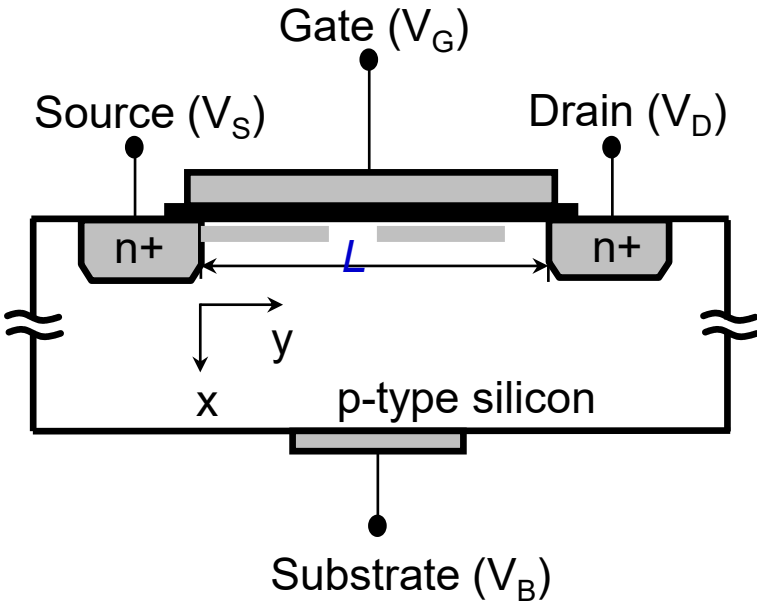
- Q_i is a function of y, since V is interchangeable with y, and Q_i is a function of V.
- Multiplying both sides of the above equation by dy and integrating from 0 to L (source to drain)

$$\int_0^L I_{DS}(y)dy = \mu_{eff} W \int_0^{V_{DS}} [-Q_i(V)]dV$$

- Rearranging the above equation, we have

$$I_{Ds} = \mu_{eff}\frac{W}{L}\int_0^{V_{ds}} [-Q_i(V)]dV$$

- Different methods have been developed with different estimations of $Q_i(y)$



Pao-Sah Model

- In this model, $Q_i(y)$ is calculated numerically by integrating the electron concentration in the x-direction
- Recall, the electron concentration is given as below

$$n(x,y) = n(\psi, V) = \frac{n_i^2}{N_a} \exp \frac{q(\psi - V)}{kT}$$

- So, $Q_i(y)$ can be calculated as below

$$Q_i(V) = -q \int_{\psi_s}^{\psi_B} n(\psi, V) \frac{dx}{d\psi} d\psi$$

$$= -q \int_{\psi_B}^{\psi_s} \frac{\frac{n_i^2}{N_A} \exp \frac{q(\psi - V)}{kT}}{E(\psi, V)} d\psi$$

where ψ_S is the surface potential at x=0 and $E(\psi,V)=-d\psi/dx$

- Thus, the drain current is

$$I_{DS} = q\mu_{eff} \frac{W}{L} \int_0^{V_{ds}} \int_{\psi_B}^{\psi_s} \frac{\frac{n_i^2}{N_A} \exp \frac{q(\psi - V)}{kT}}{E(\psi, V)} d\psi \, dV$$

- The surface potential ψ_S as a boundary value can be determined by the following equation

$$V_G = V_{FB} + \psi_s - \frac{Q_s}{C_{ox}}$$

where, the total charge induced in the silicon can be given as

$$Q_s = -\varepsilon_{Si} E_s(\psi_s)$$

- $E_s(\psi_s)$ can be obtained by letting $\psi = \psi_S$

$$E_s(\psi_s) = E(\psi = \psi_S, V)$$

Advantages:
- Valid in all device region (sub-threshold, linear, and saturation) as it includes both drift and diffusion current
- The transition region from one region to another is achieved automatically
- Its accuracy make it to be bench mark

Disadvantages
- It can be only solved numerically
- Long computation time makes it unsuitable for circuit simulation

Charge-Sheet Model

Assumption 6: Charge-Sheet Approximation

- Adopting Depletion Approximation:
  - The depletion layer under the gate is assumed to be free of mobile carriers
  - The mobile charge density Q_i is only due to inversion
- Drain current consists of drift component and diffusion component
- Drain current can be calculated as follows:

$$I_{DS}(y) = I_{drift}(y) + I_{diff}(y)$$

$$I_{DS}(y) = \mu W(-Q_i)\frac{d\psi_S}{dy} + W\mu\frac{kT}{q}\frac{dQ_i}{dy}$$

$$I_{DS} = \frac{\mu W}{L}\left[\int_{\psi_{S0}}^{\psi_{SL}}(-Q_i)d\psi_S + \frac{kT}{q}\int_{Q_{i0}}^{Q_{iL}}dQ_i\right]$$

where, ψ_{s0}, ψ_{sL}, Q_{i0}, Q_{iL} are surface potential and inversion charge at the source and drain end

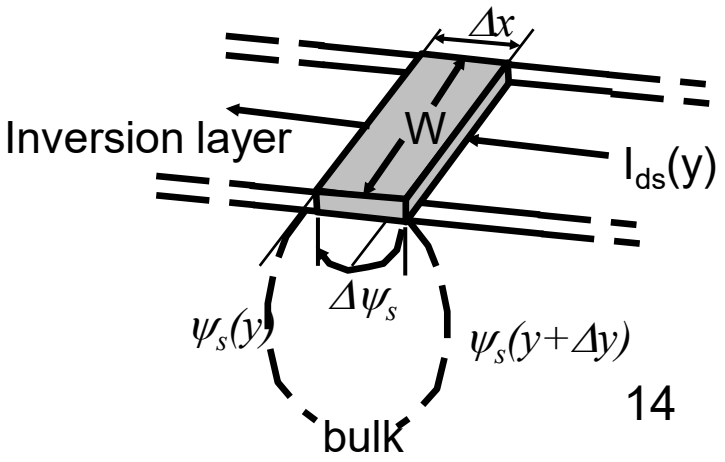
- The inversion-layer thickness is treated as zero
- All inversion charges are assumed to be located at the silicon surface
- It assumes no potential drop and no band-bending across inversion layer

$$Q_i = Q_s - Q_d$$

$$Q_s = -C_{ox}(V_{GS} - V_{FB} - \psi_S(y))$$

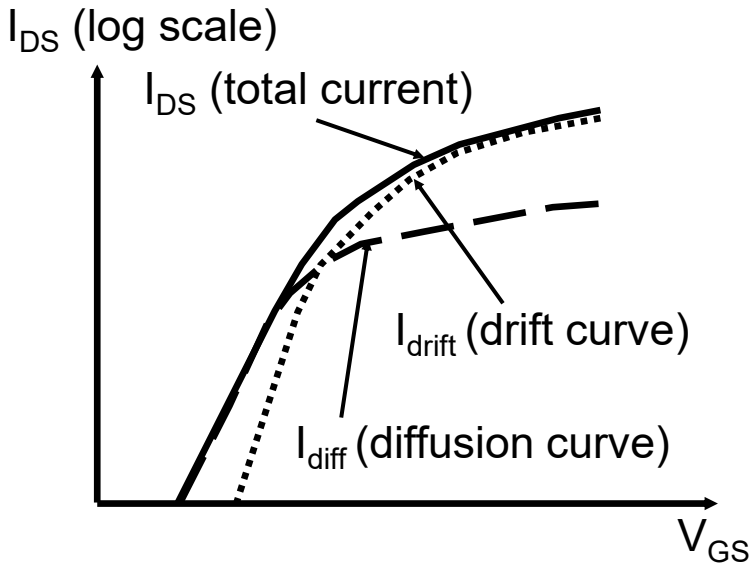
$$Q_d = -qN_A W_{dm} = -\left[2\varepsilon_{Si}qN_A\psi_S(y)\right]^{1/2}$$

$$Q_i = -C_{ox}(V_{GS} - V_{FB} - \psi_S) + \left[2\varepsilon_{Si}qN_a\psi_S\right]^{1/2}$$



Comments:

- Diffusion dominates in the weak inversion region, while drift dominates in the strong inversion region.
- The model is simpler than Pao-Sah model.
- Accuracy: worse compared to Pao-Sah model.
- But the model still consumes long time as numerical methods are required to get ψ_s .
- Large error occurs when it calculates the drain current in the weak inversion as ψ_{s0} and ψ_{sL} are very close.



Bulk Charge Model

Assumption 7: Diffusion current is negligible.

Diffusion dominates in the weak inversion region, while drift dominates in the strong inversion region, It is a good approximation for device operates in the strong inversion.

Beyond the strong inversion, surface potential remains as

$$\psi_s(y) = 2\psi_B + V(y) \qquad \psi_{s0} = 2\psi_B \text{ and } \psi_{sL} = 2\psi_B + V_{DS}$$

Total charge density in Si

$$Q_s = -C_{ox}(V_{GS} - V_{FB} - \psi_S(y)) = -C_{ox}(V_{GS} - V_{FB} - 2\psi_B - V)$$

Bulk depletion charge

$$Q_d = -qN_A W_{dm} = -\left[2\varepsilon_{Si} qN_A \psi_S(y)\right]^{1/2} = -\left[2\varepsilon_{Si} qN_A (2\psi_B + V)\right]^{1/2}$$

Inversion charge ($Q_i = Q_s - Q_d$)

$$Q_i = -C_{ox}(V_{GS} - V_{FB} - 2\psi_B - V) + \left[2\varepsilon_{Si} qN_a (2\psi_B + V)\right]^{1/2}$$

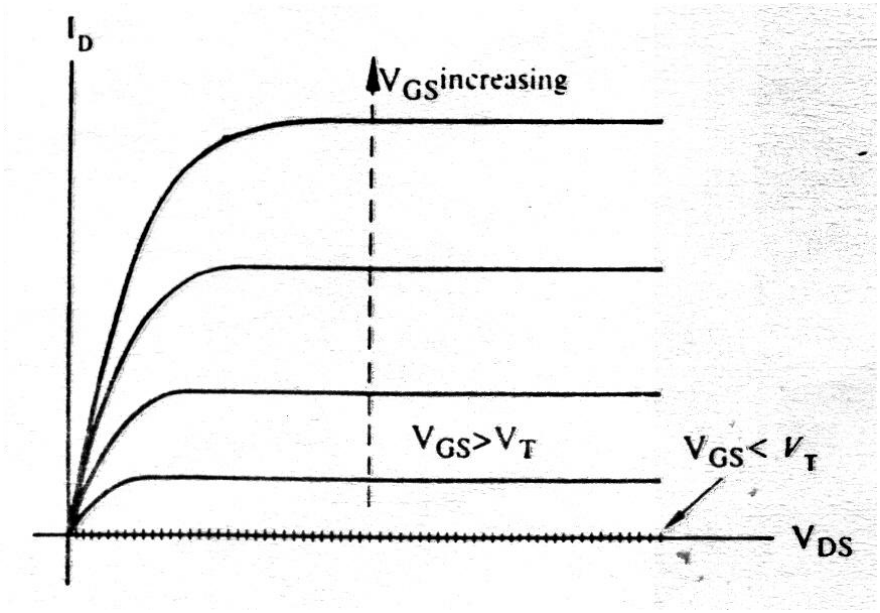
Drain current can be calculated:

$$I_{DS} = \mu_{eff} C_{ox} \frac{W}{L}\left[\left(V_g - V_{fb} - 2\psi_B - \frac{V_{ds}}{2}\right)V_{ds} - \frac{2(2\varepsilon_{Si} qN_A)^{1/2}}{3C_{ox}}\left[(2\psi_B + V_{ds})^{3/2} - (2\psi_B)^{3/2}\right]\right]$$

- For a typical MOSFET output characteristics: at a given V_{GS}, the drain current I_{DS} first increases linearly with drain voltage V_{DS} (called linear region) and then levels off gradually (called saturation region)
- The above model (equation) shows the basic output characteristics in the linear region.
- The leveling-off of the I_{DS} vs V_{DS} occurs due to pinch-off of the channel near the drain with the increase of drain voltage V_{DS}.
- Drain saturation voltage, $V_{DS,sat}=V_{DS}$ at

$$\frac{dI_d}{dV_{ds}} = 0$$

- Drain current saturates at saturation region.



Square Law Model

Assumption 8: the bulk depletion charge density Q_d is constant along the length of the channel

Under strong inversion, the surface potential fixed as

$$\psi_s(y) = 2\psi_B + V(y)$$

Bulk depletion charge $Q_d(y)$

$$Q_d(y) \approx -C_{OX}\gamma\sqrt{\psi_S(y)} = -C_{OX}\gamma\sqrt{2\psi_B}$$

$$\gamma = \frac{\sqrt{2\varepsilon_0\varepsilon_{si}qN_A}}{C_{OX}}$$

Inversion charge density $Q_i(y)$

$$Q_i(y) = -C_{OX}[V_{GS} - V_{FB} - 2\psi_B - V(y)] - Q_d(y)$$

$$Q_i(y) = -C_{OX}[V_{GS} - V_T - V(y)]$$

where,

$$V_T = V_{FB} + 2\psi_B + \gamma\sqrt{2\psi_B}$$

The drain current can be calculated:

$$I_{DS} = \mu_{eff}C_{OX}\frac{W}{L}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS}$$

Valid only when $V_{GS} > V_T$

Linear region ($V_{GS}>V_T$, $V_{DS}<V_{DS,\,sat}$)

- Process trans-conductance parameter κ:

$$\kappa = \mu_{eff} C_{OX}$$

- Gain factor β:

$$\beta = \kappa \frac{W}{L} = \mu_{eff} C_{OX} \frac{W}{L}$$

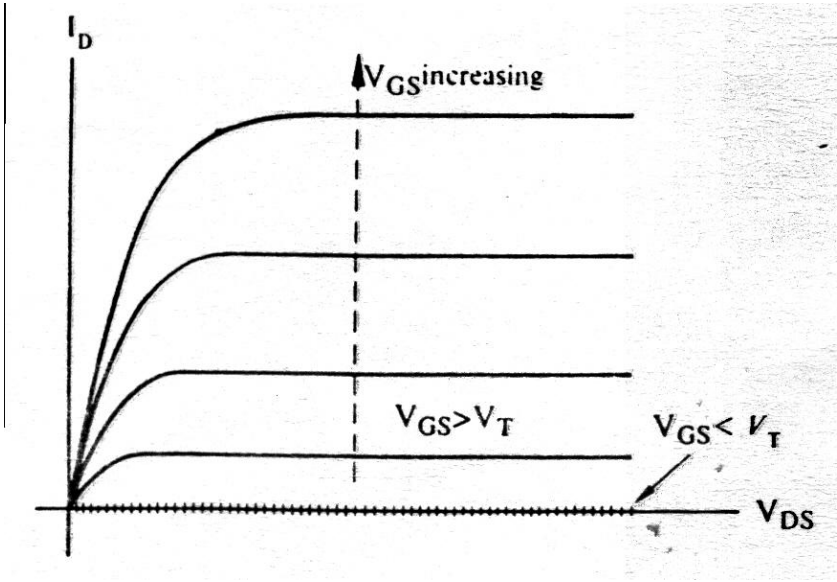
- When V_{DS} is small (< 0.1 V), drain current can approximated as:

$$I_{DS} = \beta (V_{GS} - V_T) V_{DS} \quad (V_{GS} > V_T, V_{DS} < V_{DS,sat})$$

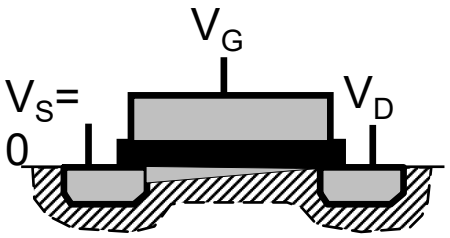
- Channel resistance

$$R_{CH} = \frac{V_{DS}}{I_{DS}} = \frac{1}{\beta} \frac{1}{V_{GS} - V_T}$$

- V_T can be extrapolated from the I_{DS}-V_{DS} plot in linear region (called linearly extrapolated threshold voltage).



Pinch-off Point ($V_{DS}=V_{DS,\,sat}$)



Pinch-off point

$V_{GS}>V_T$, $V_{DS}=V_{GS}-V_T$

- Pinch-off point (Saturation point): occurs when $V_{DS} = V_{GS} - V_T$
- Some discussions:
  - The inversion charge density decreases with the $V(y)$:

$$Q_i(y) = -C_{OX}[V_{GS} - V_T - V(y)]$$

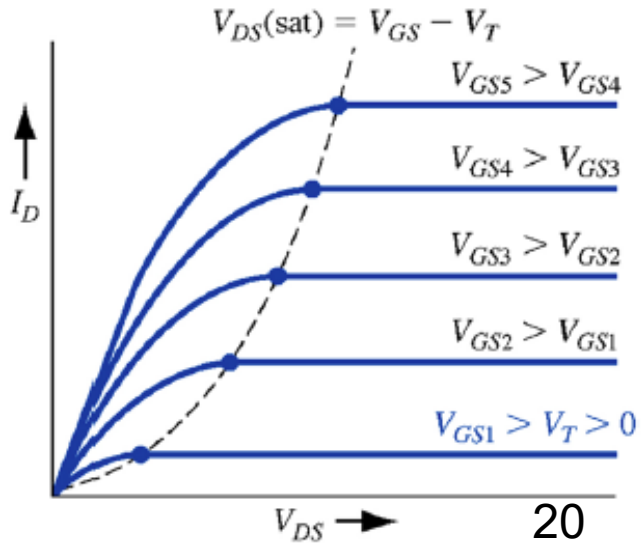
  - The inversion density Q_i at the drain end decreases as V_{DS} increases
  - When $V(y=L) = V_{DS} = V_{GS} - V_T$, the channel inversion charge density at the drain end $Q_i = 0$
  - Point-off point occurs

- Pinch-off point is equivalent to $V_{DS,sat} \equiv V_{DS} \,@\, \dfrac{dI_{DS}}{dV_{DS}} = 0$

- From the model, $V_{DS,\,sat} = V_{GS} - V_T$

- The drain current at the saturation voltage

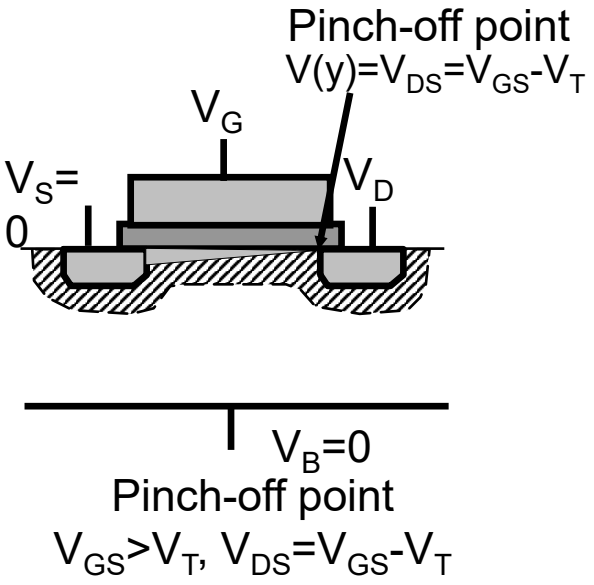
$$I_{DS,sat} = \frac{\beta}{2}(V_{GS} - V_T)^2 \quad (V_{GS} > V_T, V_{DS} = V_{DS,sat} = V_{GS} - V_T)$$

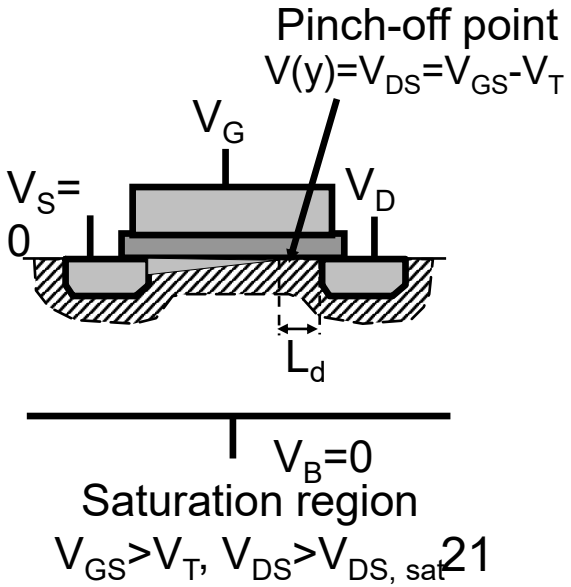


Saturation region ($V_{GS}>V_T$, $V_{DS}>V_{DS,\,sat}$)

- Further increase of V_{DS}, the pinch-off point moves towards the source end.
- Effective channel length changes from L to $\boldsymbol{L\text{-}L_d}$.
- Extra voltage greater than $V_{DS,\,sat}$ (i.e. $V_{DS} - V_{DS,\,sat}$) will be supported by the L_d depletion region.
- For long channel devices (with $L >> L_d$), the drain current will remain approximately constant at $I_{DS,\,sat}$ for any V_{DS} in excess of $V_{DS,\,sat}$
- The drain current can be calculated as following:

$$I_{DS} = I_{DS,sat} = \frac{\beta}{2}(V_{GS} - V_T)^2 \,(V_{GS} > V_T, V_{DS} \geq V_{DS,sat})$$





Summary

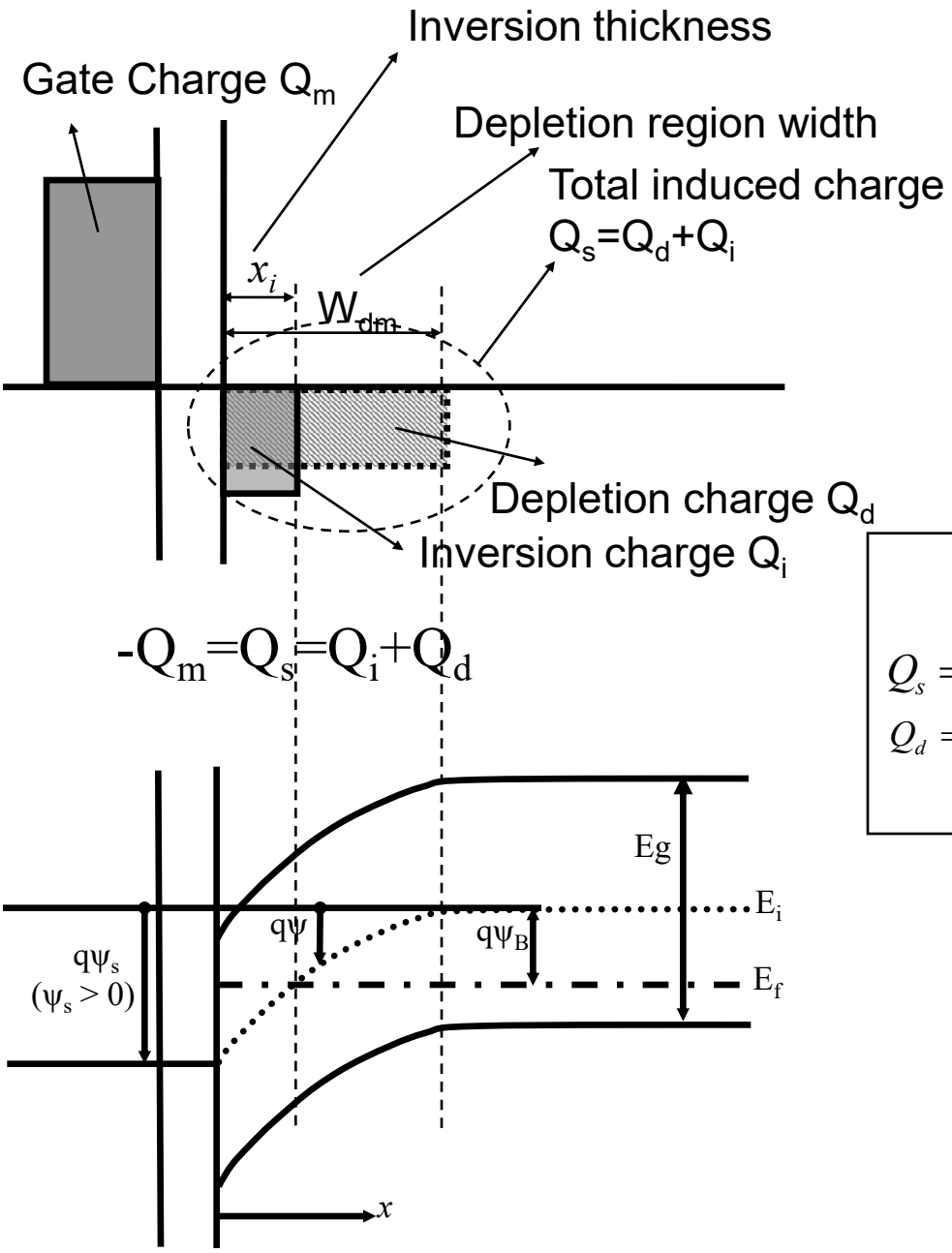
- The Square Law Model model:

$$I_{DS} = 0 \qquad (V_{GS} \le V_T)$$

$$I_{DS} = \beta(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} \qquad (V_{GS} > V_T, V_{DS} < V_{DS,sat})$$

$$I_{DS} = \frac{\beta}{2}(V_{GS} - V_T)^2 \qquad (V_{GS} > V_T, V_{DS} \ge V_{DS,sat})$$

- Following 8 assumptions are used in deriving the above model:
  1) The gradual channel approximation (GCA)
  2) Hole current can be neglected (for nMOSFET)
  3) Recombination and generation are neglected
  4) Current only flows in the y direction (along the length of channel)
  5) Carrier mobility in the inversion layer is constant
  6) Current flow is due to drift only (diffusion current is neglected)
  7) Inversion layer thickness is assumed to be zero
  8) Bulk depletion charge is constant along the y direction



Pao-Sah model

$$Q_i = -q\int_{x=0}^{x=xi} n(x,y)dx$$

Charge-Sheet Model

$$Q_i = Q_s - Q_d$$

$$Q_s = -C_{ox}(V_{GS} - V_{FB} - \psi_S(y))$$

$$Q_d = -qN_A W_{dm} = -\left[2\varepsilon_{Si} qN_A \psi_S(y)\right]^{1/2}$$

$$Q_i = -C_{ox}(V_{GS} - V_{FB} - \psi_S) + \left[2\varepsilon_{Si} qN_a \psi_S\right]^{1/2}$$

Bulk charge model

$$Q_i = Q_s - Q_d$$

$$Q_s = -C_{ox}(V_{GS} - V_{FB} - \psi_S(y)) = -C_{ox}(V_{GS} - V_{FB} - 2\psi_B - V)$$

$$Q_d = -qN_A W_{dm} = -\left[2\varepsilon_{Si} qN_A \psi_S(y)\right]^{1/2} = -\left[2\varepsilon_{Si} qN_A (2\psi_B + V)\right]^{1/2}$$

$$Q_i = -C_{ox}(V_{GS} - V_{FB} - 2\psi_B - V) + \left[2\varepsilon_{Si} qN_a (2\psi_B + V)\right]^{1/2}$$

Square law model

$$Q_i(y) = Q_s(y) - Q_d(y)$$

$$Q_s(y) = -C_{OX}[V_{GS} - V_{FB} - 2\psi_B - V(y)]$$

$$Q_d(y) = -qN_A W_{dm} \approx -(2\varepsilon_{Si} qN_a (2\psi_B))^{1/2}$$

$$Q_i(y) = -C_{OX}[V_{GS} - V_T - V(y)]$$

where,

$$V_T = V_{FB} + 2\psi_B + \gamma\sqrt{2\psi_B}$$